



Long Term Disease Treatment Monitoring and Care Coordination System

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1. Abstract

A healthcare platform to manage and monitor the treatment journey of patients with long-term and serious diseases. It organizes patients, diseases, treatment plans, medications, tests, appointments, symptoms and follow-ups, so providers can track progress and patients can view schedules and report side effects.

Main Outcome: Reliable, organized platform improving continuity of care and provider-patient coordination.

2. Problem Statement & Objectives

Problem: Long-term treatment involves many medicines, tests and visits. Scattered records make progress hard to track and follow-ups easy to miss.

- Obj 1: Design a structured database for patients, diseases and treatment plans.

- Obj 2: Build REST APIs with Node.js/Express and integrate the database with the live backend.

- Obj 3: Track medications, tests, appointments and symptom reports with notifications.

Scope: Patients and healthcare providers. **Boundary:** web-based care coordination system.

3. Existing vs Proposed System

Feature	Existing	Proposed
Treatment Records	Paper / scattered files	Centralized digital plans
Medication & Tests	Manual tracking	Scheduled, tracked over time
Appointments	Phone / manual	Booking, reminders, follow-ups
Symptoms	Reported only at visits	Patient logs anytime
Gap	Poor continuity of care → organized, coordinated platform	

4. Feasibility Study

Technical: React, Node/Express and MongoDB, all open-source and well documented.

Economic: Free tools (Git, GitHub, Postman); low development cost.

Operational: Simple dashboards for patients and providers.

Schedule: Semester project, Term-III 2025-26.

9. Results & Testing

Results

- Centralized treatment records
- Medication & test tracking
- Appointment reminders
- Symptom reporting dashboard

Testing

- REST API testing (Postman)
- CRUD operations
- Form validation
- Notification checks

Conclusion

System organizes long-term treatment data, improves continuity of care and strengthens coordination between patients and providers.

11. Future Scope

- Docker deployment & CI/CD
- Microservices split of modules
- Advanced reminders & analytics

References

- [1] React.js Docs
- [2] Node.js / Express.js Docs
- [3] MongoDB Docs
- [4] GitHub Docs

5. Tools & Technologies

Front-End: React.js, React Router, Context API / Redux

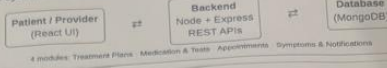
UI Styling: Tailwind CSS / Material UI

Back-End: Node.js, Express.js (REST APIs)

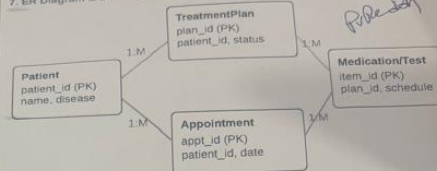
Database: MongoDB

Tools: Git, GitHub, Postman, Chart.js / Recharts

6. System Architecture

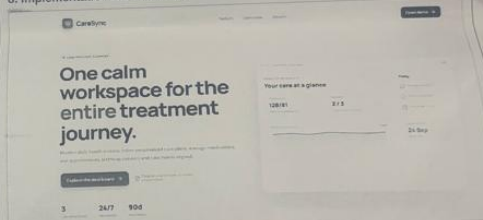


7. ER Diagram & Database Design



Also: SymptomReport (report_id, patient_id, symptom, severity, date)

8. Implementation Screenshots



Landing Page



Patient Dashboard

Reflected